

Workflow of passport-google-oauth20

Let me break down the Google authentication flow step by step using your code:

1. Initial Request (Frontend to Backend):

```
router.get("/google", ...)
```

- User clicks "Login with Google" on frontend (localhost:5173)
- Request hits `/auth/google` endpoint
- `passport.authenticate('google')` redirects to Google login page

2. Google Strategy Setup:

```
passport.use(new GoogleStrategy({...}, async (accessToken, refreshToken, profile, doneFunction) => {...}))
```

- Configures how to handle Google authentication
- Uses your Google credentials (`clientId` , `clientSecret`)
- Sets callback URL where Google will redirect after login

3. Google Login:

- User logs in on Google's page
- Google redirects back to your callback URL with user data

4. Callback Processing:

```
router.get("/google/callback", passport.authenticate('google'), googleCallbackController)
```

- Google redirects to `/auth/google/callback`
- Passport processes Google's response

5. User Processing (in Google Strategy):

```
const userEmailFromGoogle = profile.emails?.[0].value;    const userFromDatabase = await
UserModel.findOne({email: userEmailFromGoogle});
```

- Gets user email from Google profile
- Checks if user exists in database
- If not, creates new user with Google data
- Calls `doneFunction(null, userFromDatabase)` with user data

6. Session Handling:

```
passport.serializeUser((userFromGoogleStrategy, done) => {...})
```

- After successful authentication
- Stores user ID in session
- Called when user first logs in

7. Request Processing:

```
passport.deserializeUser(async (userId, done) => {...})
```

- Runs on subsequent requests
- Gets user ID from session
- Fetches full user data from database
- Attaches user to `req.user`

8. **Final Response** (in `googleCallbackController`):

```
const thisUserTokens = generateJWTAccessAndRefreshTokenFunction(userFromGoogle);
storeJwtAccessAndRefreshTokenInCookies(res, thisUserTokens); res.redirect('http://localhost:5173');
```

- Creates JWT tokens for authenticated user
- Stores tokens in cookies
- Redirects back to frontend

Flow Summary:

```
Frontend (5173)
  → Backend (/auth/google)
    → Google Login
      → Backend (/google/callback)
        → Find/Create User
          → Create JWT
            → Frontend (5173)
```

At each step, your code handles:

- Authentication state (through Passport session)
- User data (through MongoDB)
- Tokens (through JWT)
- Cookies (for token storage)